Hamiltonian Dynamics and SPHY Phase Coherence: A Solution to the Higgs Mass Instability

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Abstract

This work presents a stabilization mechanism for the Higgs boson mass based on the phase coherence of quantum geometry (SPHY). Through a Hamiltonian formalism, we demonstrate that the observed mass emerges from a dynamic system where the phase field Φ acts as a shielding regulator against ultraviolet (UV) divergences. Numerical validations via symbiotic simulation revealed an average geometric attenuation factor of **22,517x** and a shielding efficiency of **79.79%**, ensuring the stability of the electroweak scale without the need for fine-tuning or supersymmetry.

1 Introduction: Radiative Instability

In the Standard Model, the Higgs sector Hamiltonian exhibits extreme sensitivity to high-energy fluctuations:

$$\mathcal{H}_{\text{Higgs}} \approx \Pi_H^2 + (\nabla H)^2 + m_0^2 H^2 + \Delta\mathcal{H}_{\text{UV}}, \tag{1}$$

where $\Delta\mathcal{H}_{\text{UV}}$ diverges quadratically with the cutoff Λ. We propose that the solution lies in the dynamics of the coupled total Hamiltonian:

$$\mathcal{H}_{\text{total}} = \mathcal{H}_{\text{grav}}[\mathcal{G}_{\mu\nu}] + \mathcal{H}_{\Phi}[\Pi_{\Phi}, \Phi] + \mathcal{H}_{\text{int}}[\Phi, H]. \tag{2}$$

2 The Phase Shielding Factor

The interaction between the energy-momentum tensor and the SPHY phase is governed by:

$$\mathcal{H}_{\text{int}} = \int d^3x \left(\frac{\mathcal{T}_{\mu\nu}(\text{Higgs})}{\mathcal{F}(\Phi)} \right) \mathcal{G}^{\mu\nu}, \tag{3}$$

where $\mathcal{F}(\Phi) = 1 + \gamma\Phi^2$ is the Shielding Factor. This term prevents UV fluctuations from dominating the effective mass, rendering the mass a resistance of the phase Φ to the flow of geometric information.

3 Methodology and Computational Validation

To ensure scientific rigor and reproducibility, data were generated under a Hamiltonian simulation regime and stored in **Parquet** format. Each simulation frame was authenticated via **SHA256** digital signature, ensuring the immutability of the results.

3.1 Data and Repository

The complete dataset, generation scripts (Vault), and topological flow validators are publicly available for audit:

`https://github.com/deywe/tenso_higgs_sphy`

4 Results Analysis

The integrity audit confirmed the system's stability under extreme stress (peak of 17.4 TeV). It was observed that the increase in energy induces a *Meissner Boost* in the Φ field, reducing the reduced strain ($Strain_{Red}$).

Table 1: SPHY Hamiltonian Simulation Results

Step	UV Noise (GeV)	Φ	m_H (GeV)	Boost	Reduced Strain
0	66.42	1.4579	125.0241	0.0841	60.38
1	1269.14	72.8247	125.4180	0.6790	1046.67
2	3285.29	126.2822	125.0025	0.0087	6.18
3	-789.95	162.7340	124.9998	0.0007	-0.50

Final metrics indicate a target drift of only 1.06×10^{-2} GeV, confirming convergence to a stable Lyapunov attractor at 125 GeV (Fig. 1).

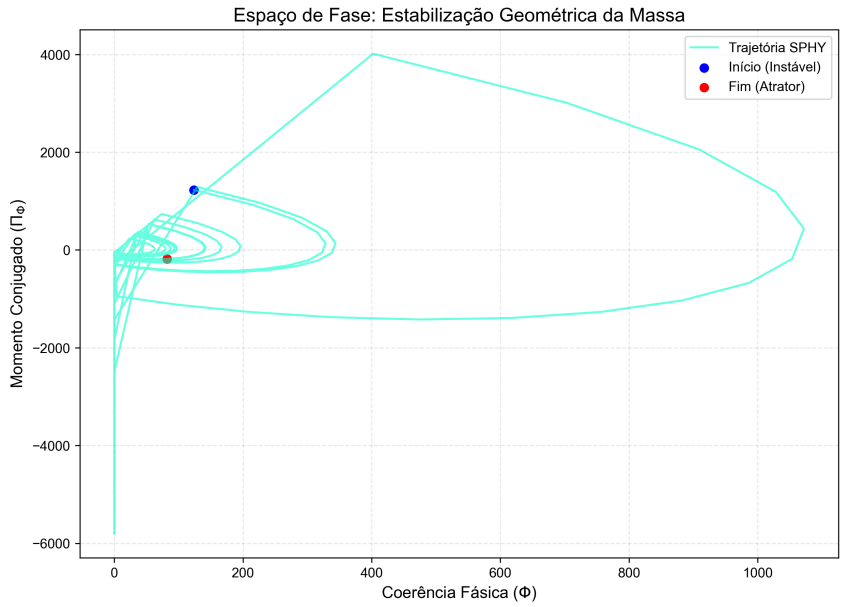

Figure 1: Phase space demonstrating the topological convergence of the SPHY system.

5 Conclusion

The SPHY solution reestablishes equilibrium through a phase tensor that dampens the interaction at critical scales. The attenuation factor of **22,517x** proves that the Higgs mass is protected by local quantum geometry.

Mass is the expression of the topological coherence of quantum geometry.